

Early Notable Models

ELMo · GPT · BERT · GPT-2 · T5 · GPT-3

Outline

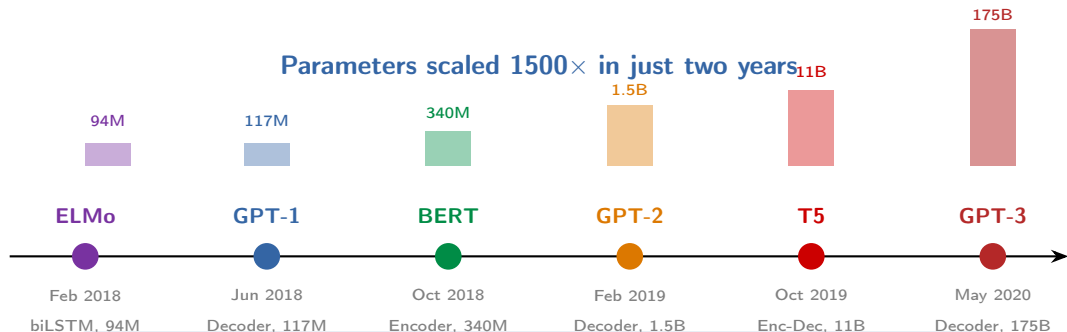
The 2018 breakthroughs

BERT

The GPT line & T5

Lessons & what came next

The timeline: 2018–2020



Each model introduced a key idea that shaped the field:
contextual embeddings → pre-train + fine-tune → bidi-
rectional → zero-shot → text-to-text → few-shot

ELMo — contextual embeddings (2018)

Before ELMo: Word2Vec / GloVe give each word a **single fixed vector**
“bank” has the same embedding in “river bank” and “bank account”

Architecture:

Character CNN →
2-layer biLSTM (4096 units)
Forward: predict next token
Backward: predict prev token
Final: learned weighted
sum of all layers

Innovation:

Same word → **different vectors**
depending on context!
Lower layers: syntax
Higher layers: semantics
Trained on 1B words

“I went to the **bank** to deposit money” → finance vector

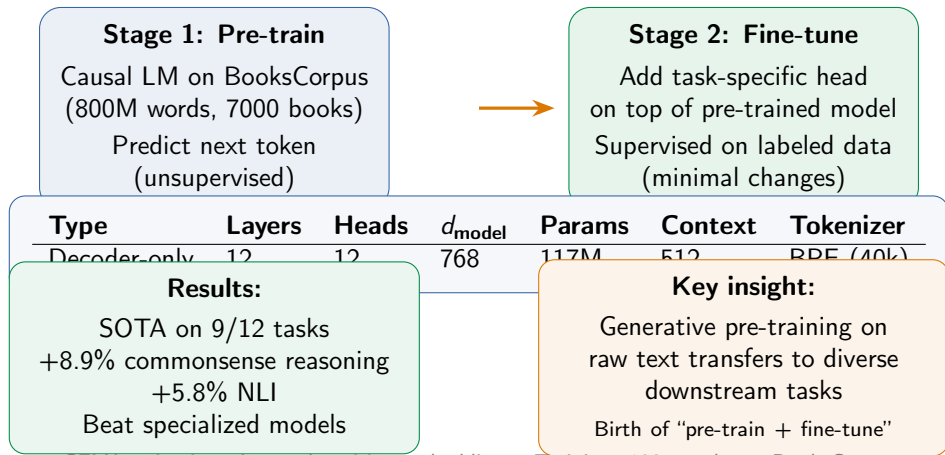
“I sat on the river **bank**” → geography vector

The *same* word gets *different* representations — this was revolutionary

Impact: proved that pre-trained contextual representations transfer to downstream tasks.
Directly inspired GPT and BERT. But ELMo used LSTMs — transformers would do it better.

GPT-1 — the pre-train + fine-tune paradigm (2018)

“Improving Language Understanding by Generative Pre-Training” (OpenAI)



GELU activation · Learned position embeddings · Training: 100 epochs on BooksCorpus

BERT — bidirectional is better (2018)

GPT's limitation: decoder-only = unidirectional (left-to-right only).
“The ____ sat on the mat” — GPT can't use right context to fill the blank!

BERT's solution: Masked Language Modeling (bidirectional)

GPT (unidirectional):

The [?] sat on the mat
← only sees “The”



BERT (bidirectional):

The [MASK] sat on the mat
← sees “The” AND
“sat on the mat”

	Layers	Hidden	Heads	Params	Vocab
BERT-Base	12	768	12	110M	30,522 (WordPiece)
BERT-Large	24	1024	16	340M	30,522 (WordPiece)

Trained on BooksCorpus (800M words) + English Wikipedia (2.5B words) = 3.3B words total

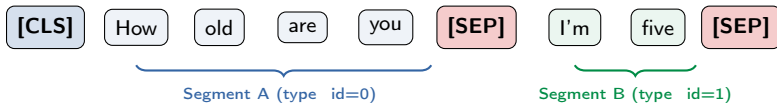
Encoder-only: no autoregressive generation. Designed for **understanding** tasks.

BERT — input format and special tokens

Single sentence:



Sentence pair:



Input = Token Embedding + Segment Embedding + Position Embedding

[CLS]

Classification token.
Its final hidden state is
the sequence
representation
for classification tasks.

[SEP]

Separator token.
Marks boundaries between
sentences. Used with
segment embeddings.

[MASK]

Masking token.
Replaces 15% of tokens
during MLM pre-training.
Never seen at
fine-tune time.

Tokenizer: WordPiece (30,522 vocab) · Subword: "playing" → "play" + "##ing" · Max length: 512

BERT — pre-training objectives

Masked LM (MLM)

Randomly mask
15% of tokens

Model predicts original token
using bidirectional context

Of the 15% selected:
80% → [MASK]
10% → random token
10% → unchanged

“The cat [MASK] on the mat”
→ predict “sat”

Next Sentence Prediction

Given sentence A and B,
predict: is B the actual next
sentence after A?

50% real pairs (IsNext)
50% random pairs (NotNext)
Binary classification on [CLS]

[CLS] A [SEP] B [SEP]
→ IsNext / NotNext

Later found: NSP was too easy (topic detection, not coherence).
RoBERTa (2019) dropped NSP entirely and got **better** results.
ALBERT replaced it with Sentence Order Prediction (SOP).

Training: 4 Cloud TPUs (16 chips), 4 days · Estimated cost: ~\$500 (2018 prices!)

BERT — why everyone lost their minds

BERT set SOTA on 11 NLP tasks simultaneously (October 2018)

GLUE benchmark:

Previous SOTA: 72.8

BERT-Large: **80.5**

+7.7 points absolute improvement

SQuAD v1.1 (QA):

Previous SOTA: 91.7 F1

BERT-Large: **93.2 F1**

First to surpass human-level (91.2)

MultiNLI:

Previous SOTA: 82.1

BERT: **86.7**

CoNLL NER:

Previous SOTA: 92.2 F1

BERT: **92.8 F1**

The BERT effect: a single pre-trained model, fine-tuned with just a linear head, beat *years* of task-specific engineering on virtually every NLU benchmark.

BERT became the default starting point for NLP from late 2018 through 2020

Spawned: RoBERTa, ALBERT, DistilBERT, DeBERTa, SpanBERT, SciBERT, BioBERT, ...

Three architectures, three strengths

Encoder-only

Bidirectional
attention

Best for:
understanding
NLI, NER, QA,
classification

BERT, RoBERTa

Decoder-only

Causal ($L \rightarrow R$)
attention

Best for:
generation
Text, code,
chat, reasoning

GPT, LLaMA, Mistral

Encoder-Decoder

Bidirectional enc +
causal dec

Best for:
seq-to-seq
Translation,
summarization

T5, BART, mBART

2024 landscape: decoder-only dominates
(GPT-4, Claude, LLaMA, Mistral, Gemini).

With enough scale + prompting, decoder-only handles understanding tasks too.

BERT-style encoders still used for: embeddings, retrieval, NER, classification at small scale

GPT-2 — scaling up, zero-shot emerges (2019)

“Language Models are Unsupervised Multitask Learners” (OpenAI)

Variant	Params	Layers	Hidden
Small	117M	12	768
Medium	345M	24	1024
Large	762M	36	1280
	1.5B	48	1600

vs. GPT-1:

13× more parameters
Pre-LayerNorm (more stable)
Context: 1024 tokens (vs. 512)
WebText: 40 GB

Zero-shot breakthrough:

SOTA on 7/8 LM benchmarks
with **no fine-tuning at all**
Coherent multi-paragraph text

“Too dangerous to release”: OpenAI withheld the full model for 9 months.

Staged release: Small (Feb) → Medium
(May) → Large (Aug) → XL (Nov 2019)

First major debate about responsible AI release. Mostly a PR event in hindsight.

Key lesson: scale alone produces qualitatively
new capabilities. Model “still underfits WebText.”

T5 — everything is text-to-text (2019)

Unified framework: cast every NLP task as “text in, text out”
Same model · same loss · same hyperparameters · different task prefixes

Translation:

“translate English to German: Hello”
→ “Hallo”

Summarization:

“summarize: [long article]”
→ “[short summary]”

Classification:

“sst2 sentence: The movie was great”
→ “positive”

Question answering:

“question: What is the capital? context: ...”
→ “Paris”

Variant	Params	Enc/Dec layers	Trained on
Small	60M	6 / 6	C4 (750 GB)
Base	220M	12 / 12	C4
Large	770M	24 / 24	C4

Encoder-decoder with span corruption objective · Introduced **C4 dataset** (750 GB cleaned web text)
Massive systematic study: compared architectures, objectives, data sizes, transfer approaches

T5 — the definitive transfer learning study

T5 paper systematically compared every design choice:

Architectures

Encoder-decoder
Decoder-only
Prefix LM

Winner: enc-dec

Objectives

Autoregressive (CLM)
BERT-style (MLM)
Span corruption
Prefix LM

Winner: span corruption

Data quality

Unfiltered web
Filtered web (C4)
Different domains

Winner: clean + large

Transfer

Fine-tune all params
Adapters
Multi-task
Gradual unfreezing

Winner: fine-tune all

Key findings:

1. Encoder-decoder > decoder-only for most tasks (at equal compute)
2. Span corruption (denoising) > standard MLM > autoregressive
3. Scaling model size + data size + training steps all help, roughly equally
4. Data quality matters more than quantity — C4's cleaning was crucial

GPT-3 — the leap to 175B (2020)

“Language Models are Few-Shot Learners” (OpenAI, NeurIPS 2020)

Layers	Heads	d_{model}	Head dim	Context	Params
96	96	12,288	128	2048	175 billion

100× larger than GPT-2, 1500× larger than GPT-1

Training data:

300B tokens total
Common Crawl (60%)
WebText2 (22%)
Books (16%) + Wikipedia (3%)

Training cost:

355 GPU-years
~**\$4.6 million**
10,000 V100 GPUs for weeks

8 model sizes spanning 3 orders of magnitude:

Ada (350M) → Babbage (1.3B) → Curie (6.7B) → **Davinci (175B)**

June 2020: OpenAI launched the GPT-3 API — first major “LLM as a service”
Text in, text out. No training needed. This began the era of API-based AI.

GPT-3 — in-context learning

No gradient updates. No fine-tuning. Just text in the prompt.

Zero-shot

Task description only.
No examples.

“Translate English to
French: cheese →”

Relies on pre-training

One-shot

1 demonstration example.

“sea otter → loutre
de mer
cheese →”

Learn format from 1 example

Few-shot

10–100 examples
in the prompt.

“sea otter → loutre
peppermint → menthe
cheese →”

Near fine-tuned quality

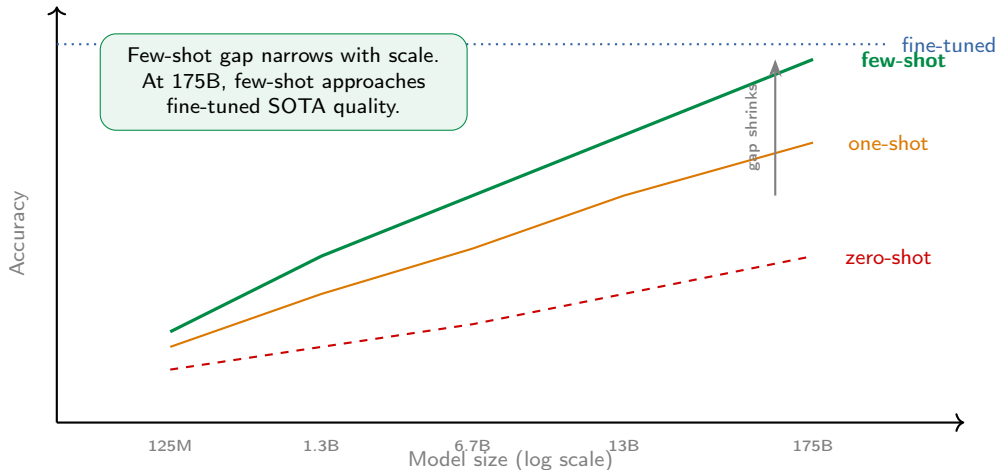
Key finding: few-shot performance **scales much faster with model size** than zero-shot. Larger models are dramatically better at learning from context.

CoQA (QA): 85.0 F1 few-shot · Competitive with fine-tuned SOTA on many benchmarks

Translation, arithmetic, code generation, common sense — all from prompting alone

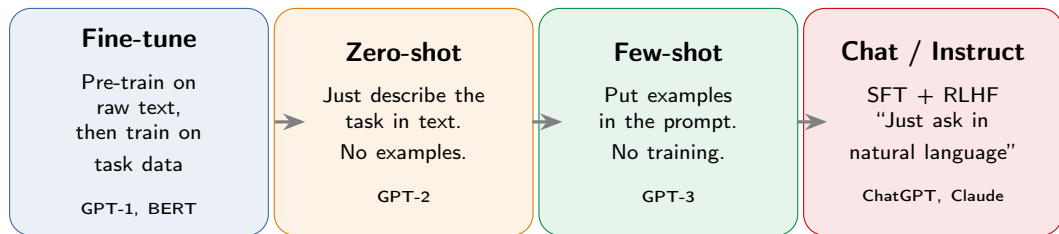
GPT-3 — few-shot scales with model size

Accuracy vs. model size on various tasks



The evolution: from fine-tuning to prompting

How we use language models evolved as they scaled



Trend: task-specific data requirements **decreased** as model scale **increased**

10k labeled examples → 100 examples → 10 examples → 0 examples (just instructions)

T5's text-to-text anticipated this: if every task is text-in/text-out, the interface naturally becomes a prompt. Scale made it work without training.

Data matters: the data scaling story

Model	Dataset	Size	Key characteristic
ELMo	1B Word Benchmark	1B words	Curated news text
GPT-1	BooksCorpus	800M words	7,000 unpublished books
BERT	Books + Wikipedia	3.3B words	Two curated sources
GPT-2	WebText	40 GB	Reddit-filtered web (3+ karma)
T5	C4	750 GB	Cleaned Common Crawl
GPT-3	Mixed (5 sources)	300B tokens	Weighted sampling strategy
LLaMA	Mixed (8 sources)	1.4T tokens	Publicly available data only
LLaMA 2	Unknown	2T tokens	Undisclosed mix

Quality > quantity:

T5 showed filtered C4 beats unfiltered Common Crawl.
GPT-3 upsampled high-quality sources (Wikipedia 3×).

Diversity matters:

GPT-3 mixed: web, books, Wikipedia, code.
Broad coverage → broader capabilities.

From 1B words (2018) to 2T tokens (2023) — a 2000× increase in training data

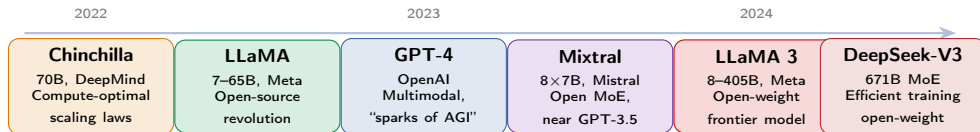
Summary: what each model taught us

Model	Year	Architecture	Params	Key contribution
ELMo	2018	biLSTM	94M	Contextual embeddings (same word $\rightarrow$ different vectors)
GPT-1	2018	Decoder	117M	Pre-train + fine-tune paradigm
BERT	2018	Encoder	340M	Bidirectional (MLM), SOTA on 11 tasks
GPT-2	2019	Decoder	1.5B	Zero-shot emerges from scale
T5	2019	Enc-Dec	11B	Text-to-text unification, systematic study
GPT-3	2020	Decoder	175B	Few-shot in-context learning, API launch

The themes that emerged:

1. Scale unlocks new capabilities (GPT-1 $\rightarrow$ 2 $\rightarrow$ 3)
2. Pre-training on raw text transfers to any task (GPT-1, BERT)
3. Architecture choice matters less than scale (T5 study)
4. Data quality is as important as quantity (T5, GPT-3)
5. Prompting can replace fine-tuning at sufficient scale (GPT-3)

What came next: the post-2020 landscape



Key themes of the post-GPT-3 era

Compute-optimal scaling

Chinchilla showed models were over-parameterized & under-trained. Optimal: scale data & params equally.

Open-source/weight

LLaMA sparked an explosion of open models: Alpaca, Vicuna, Mistral, Phi, Qwen, ...

Multimodal + reasoning

GPT-4, Gemini, Claude: vision, audio, tool use, extended thinking

Mixture of Experts

Mixtral, DeepSeek-V3: large capacity with efficient inference via sparse activation

We will cover these models, RLHF/alignment, and advanced prompting in upcoming lectures.

Further reading

GPT Series

- Radford et al. (2018), “Improving Language Understanding by Generative Pre-Training” (GPT-1)
- Radford et al. (2019), “Language Models are Unsupervised Multitask Learners” (GPT-2)

Devlin et al. (2019), “BERT: Pre-training of Deep Bidirectional Transformers”

BERT & Variants

- Devlin et al. (2019), “BERT: Pre-training of Deep Bidirectional Transformers”
- Liu et al. (2019), “RoBERTa: A Robustly Optimized BERT Pretraining Approach”

T5 & Encoder-Decoder

- Raffel et al. (2020), “Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer” (T5)
- Lewis et al. (2020), “BART: Denoising Sequence-to-Sequence Pre-training”

Questions?

Next: Prompting — Zero-shot, Few-shot, Chain-of-Thought